

Dyuman Aditya

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🔗 Google Scholar

in dyumanaditya

🌐 <https://dyumanaditya.github.io/>



Education

- 2026 – Pres. 📖 **Incoming CS PhD at University of Southern California**
- 2023 – 2025 📖 **M.Sc. Ecole Centrale de Nantes** Advanced Robotics: Control & Robotics (class rank 1). Thesis title: *Bundled Contacts: Learning transferable policies in differentiable simulation*.
- 2019 – 2022 📖 **B.Sc. Sri Aurobindo International Center of Education** in Computer Science, Mathematics (class rank 1).

Research Publications

(* denotes equal contribution)

Conference Proceedings

- 1 **D. Aditya**, J. Huang, N. Bohlinger, *et al.*, “Robust localization, mapping, and navigation for quadruped robots,” in *ECMR*, (Oral presentation), 2025.
- 2 **D. Aditya**, C. Payne, M. Levia, and P. Shakarian, “Machine learning model integration with open world temporal logic for process automation,” in *ICLP*, (Oral presentation), 2025.
- 3 D. Bavikadi, **D. Aditya**, D. Parkar, *et al.*, “Geospatial trajectory generation via efficient abduction: Deployment for independent testing,” in *ICLP*, (Oral presentation), 2024.
- 4 **D. Aditya***, K. Mukerji*, S. Balasubramanian, A. Chaudhary, and P. Shakarian, “PyReason: Software for open world temporal logic,” in *AAAI Spring Symposium*, (Oral presentation), 2023.
- 5 K. Mukerji, D. Parkar, L. Pokala, **D. Aditya**, and P. Shakarian, “Scalable semantic non-markovian simulation proxy for reinforcement learning,” in *IEEE ICSC*, (Oral presentation), 2023.

Journal Articles

- 1 K. Mukherji, J. M. Patil, **D. Aditya**, *et al.*, “Lattice annotated temporal (lat) logic for non-markovian reasoning,” *ACM*, 2025, Under review, submitted.

Patents

- 1 P. Shakarian, **D. Aditya**, and K. Mukherji, “Systems and methods for open world temporal logic,” Patent application US App. 19/044,380, Aug. 7, 2025.
- 2 S. Pal, K. Chakraborty, **D. Aditya**, A. Datta, and J. Peters, “Control of industrial machines by tracking the movements of their operators,” 2021.

Awards

- 2025 📖 ETH Zurich Robotics Student Fellowship (3%)
- 2022 📖 Best Student Award, Sri Aurobindo International Center of Education (1%)

Work Experience

1 2025 Feb–Present **Visiting Student Researcher**
Computational Robotics lab ETH Zurich (MSc. Thesis), Zürich

- Worked on a novel method called "Bundled Contacts" to train transferable robot policies in differentiable simulation.
- Working on converting my thesis into a research publication.

2 2024 July–Present **Research Assistant**
IAS Lab TU Darmstadt

- First authored a paper on exploiting the kinematics of low-cost quadruped robots for robust localization and mapping.
- Co-author on a project involving contact localization with hierarchical flow matching.

3 2022 March–2025 October **Research Associate**
Leibniz Lab, Syracuse University

- Worked on multiple projects (see publications) in the field of Neuro Symbolic AI
- First authored two manuscripts and Co-authored three others
- Developed and maintained PyReason, a high-performance software for open-world temporal logic.
- Developed integration methods for Machine learning models into PyReason for Robotic welding scenarios

4 2023 June–2024 September **Research Assistant**
ARLIS, University of Maryland

- Worked with the US Air Force Research Laboratory (AFRL) to develop military operational workflow verification using PyReason as a simulation.

5 2022 June–2022 November **Internship**
Xu Lab, Carnegie Mellon University

- Worked on evaluating the performance of algorithms that determine the position and orientation in biological images (Tomograms)

6 2020 July–2022 August **Robotics and Machine Learning Engineer**
Telekinesis AI, TU Darmstadt

- One of the first members of Telekinesis - a startup aimed at providing the industry with no-code robot programming solutions.
- Co-author on the first company patent (see patents)
- Developed the company's core software pipeline, used till date
- Built a modular Reinforcement Learning toolkit with state-of-the-art algorithms
- Built multiple environments with PyBullet: Bin-picking, Drilling etc.
- Developed the front end visualization software for interfacing with real robots